

MODULE 4

Envisioning Future Possibilities

Methods for Envisioning Future Possibilities: Concept Ideation, Modeling and Prototyping, Design Rationale.

Design Thinking, Ideation, and Sketching: Design paradigms, Design thinking, Design perspectives, User personas, Ideation, Sketching.

Methods for Envisioning Future Possibilities

- Envisioning future possibilities in design is a **critical step that enables designers to think beyond current constraints and anticipate innovative solutions to user problems.**
- This process involves:
 - **concept ideation** (where many possible ideas are generated),
 - **modeling and prototyping** (where these ideas are converted into simple models or prototypes to understand how they might work), and
 - creating a **design rationale** (record of the reasons behind the decisions made during the design process).
- These methods help translate initial insights into tangible ideas and possibilities that shape the direction of the design.

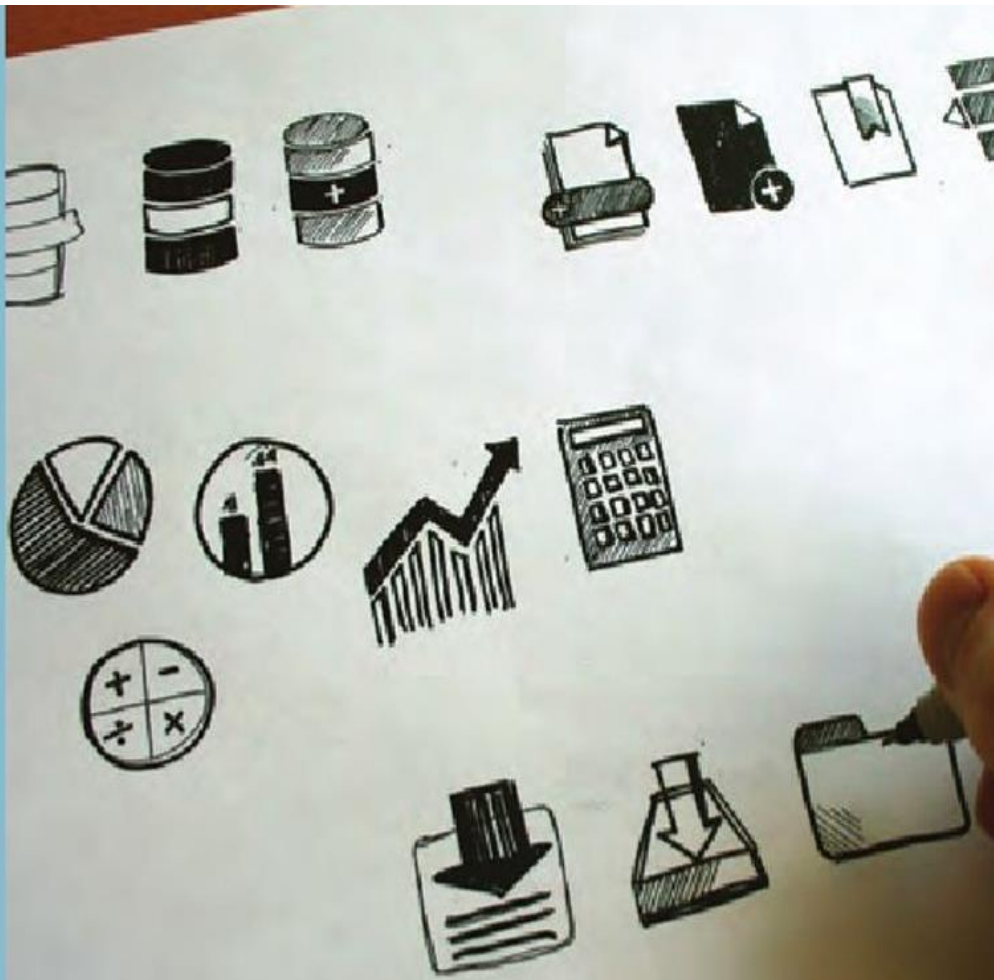
Concept Ideation

- **Concept ideation** is the process of generating multiple ideas and solutions for addressing design challenges.
- It encourages creativity and exploration of diverse possibilities.
- The goal is to imagine a variety of potential solutions, regardless of feasibility at this stage, to ensure that the best and most innovative ideas are considered.
- Some of the methods for concept ideation are:
 - Thumbnail Sketching
 - Creative Matrix
 - Round Robin
 - Alternative Worlds

Thumbnail Sketching

- A series of small drawings used to quickly explore a variety of ideas.
- The goal is **not perfection**, but to quickly visualize different possibilities.
- Designers use them to think creatively, compare options, and refine concepts before selecting the best idea for further development.
- The act of sketching promotes a natural cycle of creativity that supports concept ideation—you externalize what's in your mind, and then, upon seeing your idea visualized, you think of variants, and so on.
- The goal is simply to get the ideas out of your head and onto paper quickly, thumbnail sketches should be small, gestural, and general.

Here's an example of sketches being generated for a financial services application. The team explored various possibilities for a set of symbolic icons.



BENEFITS

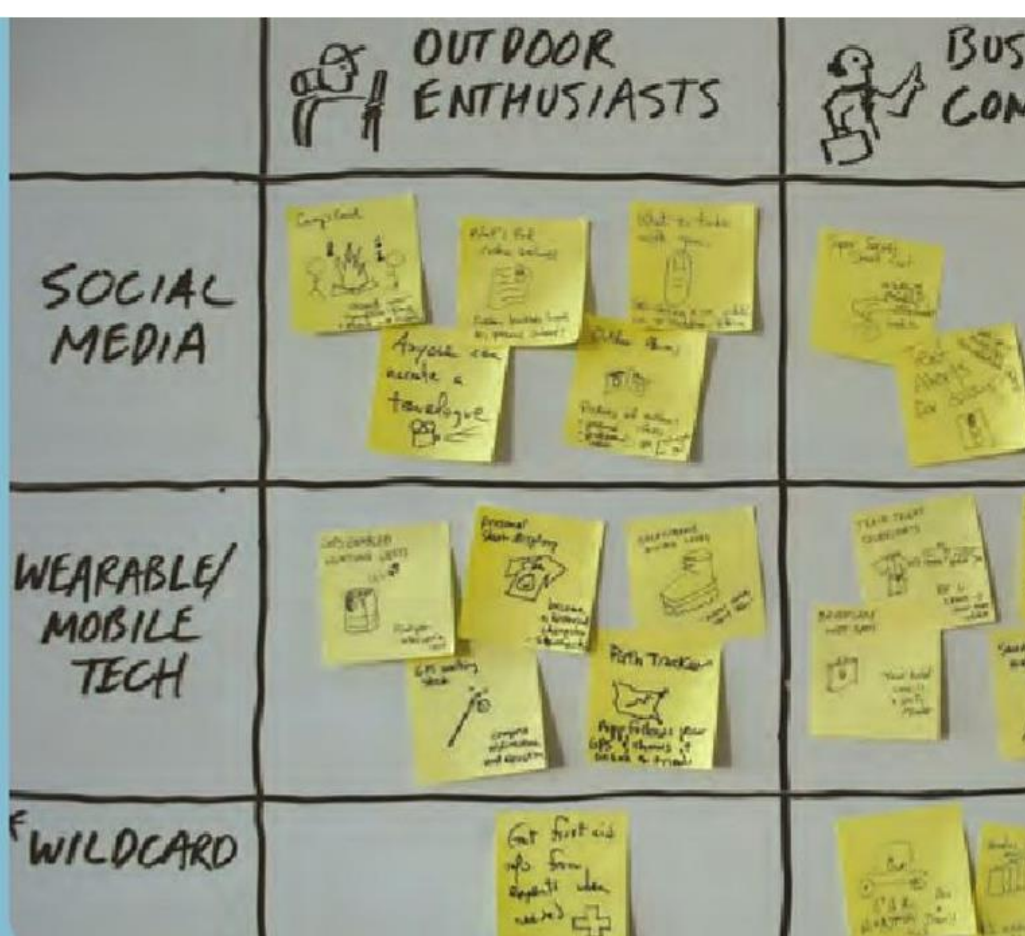
- Helps you generate ideas quickly
- Promotes divergent thinking
- Encourages sharing of ideas
- Helps you make iterative improvements

- Identify a topic for exploration.
- Assemble a team and some basic drawing materials.
- Ask each person to work independently.
- Invite them to think about one aspect of the topic.
- Ask them to draw pictures of various possibilities.
- Invite them to consider another aspect of the topic.
- Ask them to draw pictures of various possibilities.
- Invite a group discussion of the sketches.
- Pick a subset to refine with additional sketching.

Creative Matrix

- A format for sparking new ideas at the intersections of distinct categories.
- When a problem is too broad, or you're having trouble breaking away from conventional thinking, structure can actually help.
- To generate many wide-ranging ideas in a short amount of time, the structure of a Creative Matrix is useful because it stimulates cross-pollination by providing a template for generating new ideas where topics intersect.
- In a Creative Matrix grid, each cell represents the intersection of two disparate categories.
- It's best to use the columns as categories related to people (e.g., personas, market problem statements, key challenges) and the rows as categories for enabling solutions (e.g., technologies, environments, or policies).
- You can then use this “mash-up” of categories to help generate a wide range of concepts in each cell.

In this project, a company specializing in traditional luggage wanted concepts for new products and services. The design team generated 107 ideas in 10 minutes.



BENEFITS

- Helps you generate a large number of ideas
- Promotes divergent thinking
- Helps you think of new and unusual ideas
- Invites input from all team members

- Identify a design challenge in need of fresh ideas. Make a poster showing a large grid (max. 5×5 cells).
- Designate columns: Categories related to people.
- Designate rows: Categories for enabling solutions.
- Form teams.
- Hand out grids.
- Introduce the topic.
- Give each participant a pen and a sticky note pad.
- Ask them to ideate at the intersections of the grid.
- Instruct them to write one idea per sticky note.
- Start the clock.
- Limit the time to 15-20 minutes.

Eg: improving a coffee shop experience

Customer Need	Technology	Ambiance	Service
Faster Ordering	Mobile app pre-orders	Self-service kiosks	Extra staff during peak hours
Better Coffee Quality	Smart espresso machines	Cozy seating & lighting	Trained baristas
Social Experience	Free Wi-Fi & charging stations	Live music nights	Friendly, engaging staff

- This structure helps in brainstorming improvements by mixing different aspects of the business.

Round Robin

- An activity in which ideas evolve as they are passed from person to person.
- Round Robin allows for the generation of fresh ideas by providing a format for group authorship.
- As an idea is passed from person to person, it can grow and change in unexpected ways to uncover some wonderfully original concepts.
- Round Robin is a superb way to get a small team of people to create a collection of a typical ideas around a particular challenge, because it requires that participants inherit and build upon each other's ideas.
- The power of this method is that ideas emerge from collective input—everyone takes a turn.
- As you inherit another's idea, think critically about what you're given but don't let it limit your own thinking.
- Even if the idea seems strange or impossible, it may contain the seed of a successful conceptual direction.
- The best result is a set of ideas that no single person could have imagined on her own.

This example shows a team working on a new marketing campaign for an international book and music retailer. Their concepts improved with each pass of the paper.



BENEFITS

- Invites input from all team members.
- Diminishes overbearing opinions.
- Facilitates group authorship.
- Helps you think of new and unusual ideas.

- Identify a design challenge in need of fresh ideas.
- Make a worksheet folded into four parts: problem, solution, critique, resolution.
- Form teams with 4-5 people.
- Hand out worksheets.
- Instruct each person to write down the challenge.
- Ask them to write down an unconventional solution.
- Instruct everyone to pass each worksheet to the left.
- Ask them to write a reason why the proposal will fail.
- Instruct everyone to pass each worksheet again.
- Ask them to write down a way to resolve the critique.

Alternative Worlds

- A way of using different perspectives to help generate fresh ideas.
- Good designers borrow, adapt, and remix ideas from other domains.
- Thinking only within your own field can limit creativity. Looking at how others work helps you break routine thinking.
- Alternative Worlds challenges you to investigate how a successful organization or group outside your domain would approach a given issue.
- For example, if your team were tasked with reinvigorating a public library you might ask: What would Starbucks do? Doing so requires a deep understanding of their culture, practices, and procedures, in order to determine what aspects can be appropriated for your purposes.
- Adopting the perspective of a completely different domain is a novel way to shift perspective and uncover potentially breakthrough ideas.

In this example, a team that was redesigning hospitals decided to study the hospitality experience at hotels. The exercise sparked many new ideas for color and lighting.



BENEFITS

- Helps you see things from a fresh perspective.
 - Provides inspiration.
 - Leverages success from other domains
 - Helps you think of new and unusual ideas
- Identify a topic for exploration.
 - Assemble a diverse group of collaborators (designers, doctors, etc.)
 - Create a list of similar topics (hospital, hotel, resort, etc..)
 - Think about representative organizations (hotels known for comfort and service).
 - Select a few alternative worlds to explore.
 - Orient the team to key attributes of each world.
 - If possible, fully immerse them in each world.
 - Ask: How would ____ solve our challenge?
 - Seek various ideas inspired by the new perspective.

MODELING & PROTOTYPING

- This step allows designers to visualize how their ideas would function in practice and test them in real-world scenarios.
- **Modeling**
 - Involves creating representations of ideas, whether conceptual or non functional.
 - The goal is to visualize how the design will work and uncover potential problems early.
- **Prototyping**
 - It is about creating **interactive models** or **functional versions** of the design to test with users.
 - Prototypes allow users to interact with a simplified version of the design to validate ideas, uncover usability issues, and provide feedback.

- Some of the methods are:
 - Storyboarding
 - Schematic Diagramming
 - Rough & Ready Prototyping
 - Appearance Modeling

Storyboarding

- A series of images showing the key elements and interactions of a new scenario.
- THE POWER OF a good story lies in its ability to capture people's imagination, and take them to new places.
- Stories give action to ideas and place them in a context of use. Therefore, Storyboarding is an ideal way to visualize people experiencing your new idea in action.
- Since storyboards are built using images, they can help teams communicate ideas more clearly than words.
- Storyboards just need to convey a meaningful sequence of events.
- Even primitive drawings can help you envision the possibilities of new interactive, cinematic, or transactional experiences.

This example shows a new concept for home automation. It was made to tell the story of a first-time buyer — from in-store purchase to successful installation at home.



BENEFITS

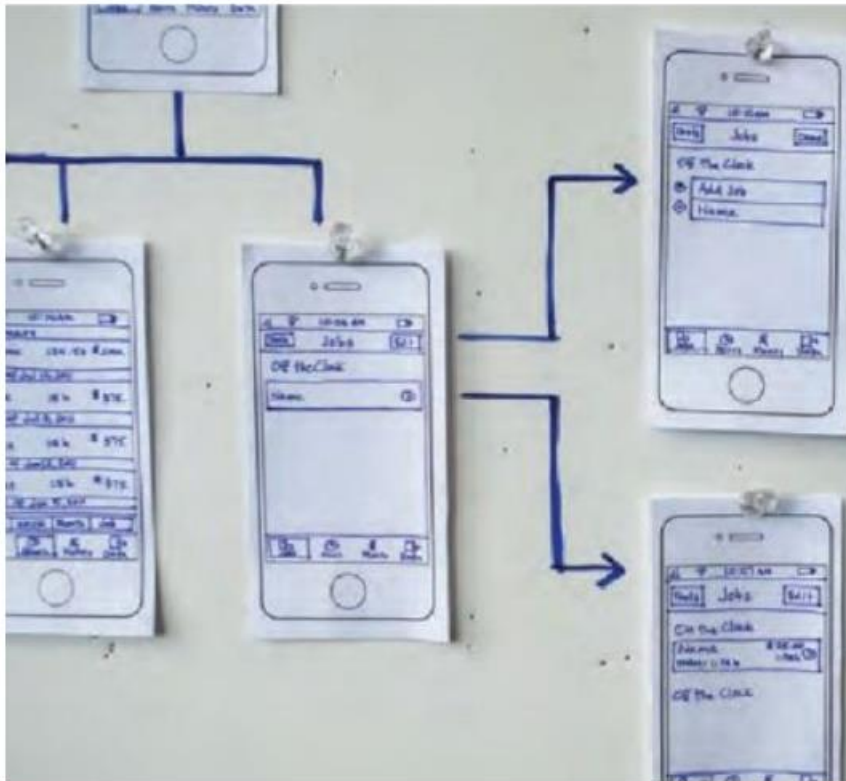
- Shows what a concept looks like in action
- Helps people imagine the future
- Builds a shared understanding
- Gains support from decision makers

- Identify an idea or concept to develop.
- **Make a poster containing 10-12 blank rectangles.**
- Assemble a diverse group of collaborators.
- Equip them with a set of basic drawing materials.
- Draft the main story line (beginning, middle, end).
- Determine the main characters and the setting.
- Draw the key frames for a future scenario.
- Include a **descriptive phrase** beneath each frame.

Schematic Diagramming

- An outline of the structure and essential components of a system.
- A schematic diagram is as an x-ray of the skeleton of your concept.
- It's the structure on which all the other components will depend, supporting their function and flow.
- A schematic diagram allows your team to collaborate on how the idea will work as a functional system.
- Don't focus on refinement here; instead, experiment with the best arrangement of parts to determine the ideal system.
- Using symbolic drawings, you need only show the compositional makeup of the basic parts and how they interact with or transition from one element to the next.
- Although it may sound technical, Schematic Diagramming can be used in nearly any project, from structuring screen layouts, to coordinating navigation, to sequencing the steps in a service experience.

In this project, a development team outlined the screen layouts and navigational flow of a new smartphone app. The diagram served as the plan for their first prototype.



BENEFITS

- Shows the structure of a proposed solution
- Helps you work out functional details
- Builds a shared understanding
- Informs subsequent design activities

- Identify an idea or concept to develop.
- Assemble a diverse group of collaborators.
- Equip them with a set of basic drawing materials.
- Determine the basic elements to be included.
- Render each element with stylistic simplicity.
- Compose the elements in a clean and clear way.
- Adjust the size and weight of elements for emphasis.
- Arrange the overall diagram in an orderly manner.

Rough & Ready Prototyping

- A rapidly built model of a new idea that approximates its appearance and behavior.
- MODEL MAKING is the fundamental craft of design—it is the process of thinking with your hands.
- Low-fidelity sketch models make it easy to generate a number of tangible concept ideas without significant time investment, allowing you to bring form to even nascent ideas.
- Using readily available, low-cost materials such as paper, markers, cardboard, glue, and so forth, you can quickly create concept models for people to evaluate.
- The sooner you create something tangible for people to touch and manipulate, the earlier you'll get meaningful feedback about your ideas.
- You can use rough models to help identify ideas that are working, as well as pinpoint weaknesses early in the process so they can be addressed before the concept is finalized.

In this example, basic materials were used to quickly consider how the information displayed on a handheld unit might affect the design of a new medical service.



- Identify an idea or concept to develop.
- Assemble a small design team.
- Gather basic materials (e.g., pen, paper, tape, etc).
- Consider what you want to learn from the prototype.
- Select a scenario and a few key tasks to focus on.
- Build a rough approximation of the concept.
- Simulate as much functionality as possible.
- Include some realistic and readable content.
- Provide a signal for the areas that are incomplete.

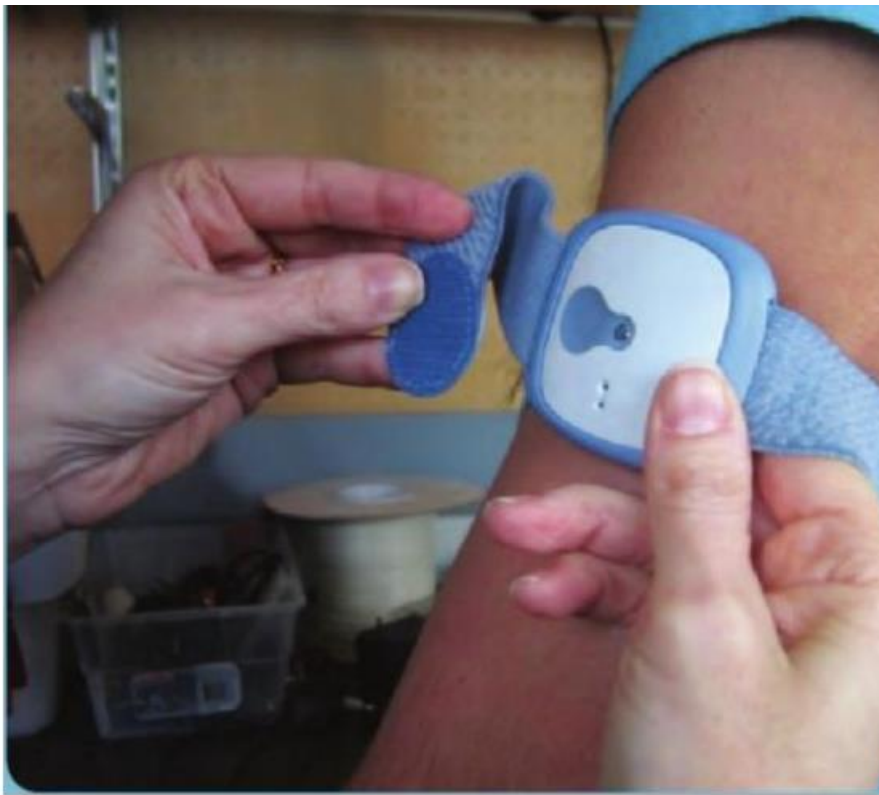
BENEFITS

- Promotes a shared vision of the future
- Helps you test ideas quickly
- Lowers development costs through early discovery
- Helps you make iterative improvements

Appearance Modeling

- A refined model of a new idea that emphasizes the visual styling.
- Functionality is important, but the way something looks and feels is essential to the success or failure of your design.
- In fact, many studies have shown that people report higher degrees of satisfaction for aesthetically appealing products and services.
- A great way to consider the visual and emotional attributes of your proposed solution is to build a model that appears to be the real thing.
- An Appearance Model is a pre-production prototype that clearly represents the look and feel of your design, but has little or no functionality.
- This type of modeling is equally useful in the development of physical products, software systems, etc..
- It is relevant to every type of solution that has any sort of experiential element.
- If you make an appearance model of a key element, you can determine if the aesthetics lead to a high degree of satisfaction.

Here's an example of a designer modeling a wearable health monitor. She is exploring the emotional and stylistic qualities of the different materials and finishes.



BENEFITS

- Helps you consider aesthetics.
 - Reveals emotional qualities.
 - Provides a vision of the future.
 - Gains support from decision makers.
- Identify an idea or concept to develop.
 - Assemble a small design team.
 - Decide what you want to learn from the model.
 - Consider a range of visual and emotional qualities.
 - Assemble a palette of colors and/or sample materials.
 - Develop sketches to show a number of possibilities.
 - Narrowing down and improving key ideas.
 - Draw realistic renderings of the concept.
 - Produce models showing the finished effect.

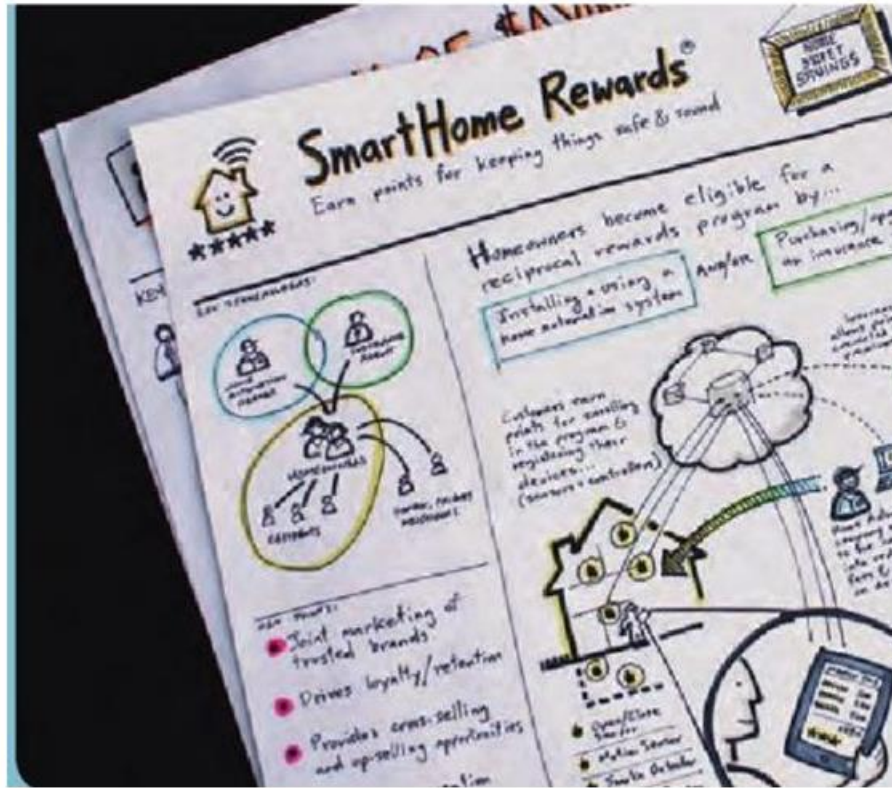
DESIGN RATIONALE

- **Design rationale** refers to the reasoning behind design decisions.
- It explains why certain design choices were made over others and justifies the design solutions based on user needs, technical constraints, and business goals.
- Creating a clear design rationale ensures that everyone involved in the project understands the intent behind the design and its alignment with user requirements and project goals.
- Unfortunately, many good ideas fail to become innovations simply because their creators fail to convey the concept's full potential.
- The methods in this grouping help you compel others to take the necessary actions for great ideas to flourish.
 - Concept Poster
 - Video Scenario
 - Cover Story Mock-up
 - Quick Reference Guide

Concept Poster

- A presentation format illustrating the main points of a new idea.
- An effective Concept Poster is a powerful way to promote an idea and rally support for its development.
- Start by thinking big—large scale, big ideas, short phrases—and then fill in the details to help support the main points you're highlighting.
- Your poster should show what the idea is, why it matters, and how it works, to influence people to embrace your concept by communicating what makes it an appropriate solution.
- Since this is a poster, you'll want to keep it highly visual, using images and words that clearly and concisely articulate the important aspects of your concept.
- Remember that while words can aid understanding, illustrations, even if rudimentarily drawn, can go a long way in conveying an idea.

In this project, a team that was working on a new insurance service illustrated the people and products involved in their concept for a rewards program.



BENEFITS

- Promotes a vision of the future
- Helps you build a business case
- Gains support from decision makers
- Provides a road map for moving forward

- Identify a new idea to introduce.
- Assemble a team and some basic drawing materials.
- Come up with a name and a tag line for the concept.
- Write a short summary of the big idea.
- Include a description of the key stakeholders.
- List a few features and benefits.
- Illustrate the concept with a big picture or diagram.
- Add a timeline for developing the solution.
- Draft the layout and draw the final poster.

Video Scenario

- A short movie showing the attributes of a new concept in use.
- IF A PICTURE is worth a thousand words, then a video is worth a thousand pictures.
- You can spend great time and effort telling people about a new concept, but words only go so far—sometimes you have to show people the world you envision.
- With video, you can present a compelling explanation of what your concept would be like in a true-to-life scenario.
- This allows people to better imagine how one might interact with or experience your design.
- As a vehicle for storytelling, a Video Scenario has the potential to capture your team's attention and gather their support for an idea before it is made real.
- These days, creating, editing, and sharing video is common place and relatively easy.
- While you may want someone with experience to polish the final product, you can do much of the groundwork yourself.
- Use your storyboarding skills to frame the narrative, and then use video to bring it to life!

Here's an example of a video being created to describe a new concept for a digital pen. Simple animation effects were used to show a rough prototype in action.



BENEFITS

- Helps people imagine the future
- Shows what a concept looks like in action
- Gains support from decision makers
- Inspires your team.

- Identify a new concept to represent.
- Assemble a small design team.
- Create a storyboard of a future scenario.
- Write a detailed script.
- Collect the prototypes, people, and other props.
- Record video or take still photos of each scene.
- Edit the footage to a short duration (2–5 minutes).
- Record and overlay narration.
- Post the final movie online for convenient viewing.

Cover Story Mock-up

- A mock news article describing the successful future of a new idea.
- It presents the idea as if it has already been successfully implemented and is being featured as a cover story in a leading publication.
- An idealized story is a unique way to explain the need or benefit of something by imagining the impact of its creation.
- If you are able to present your concept as a success, by showing its potential merits and a positive outcome, you are more likely to get buy-in from those who are initially uncertain about it.
- Your goal for a Cover Story Mock-up is to answer the question: What will this ____ have done?
- This requires you to consider the potential cultural impact of your idea and to frame its success around how people would ideally respond to it.
- When you begin with a positive ending, it's easier to regroup people around the development of your idea.

In this example, a team wrote and illustrated a story about the successful launch of a new public service for delivering farm-fresh foods to urban neighborhoods.



BENEFITS

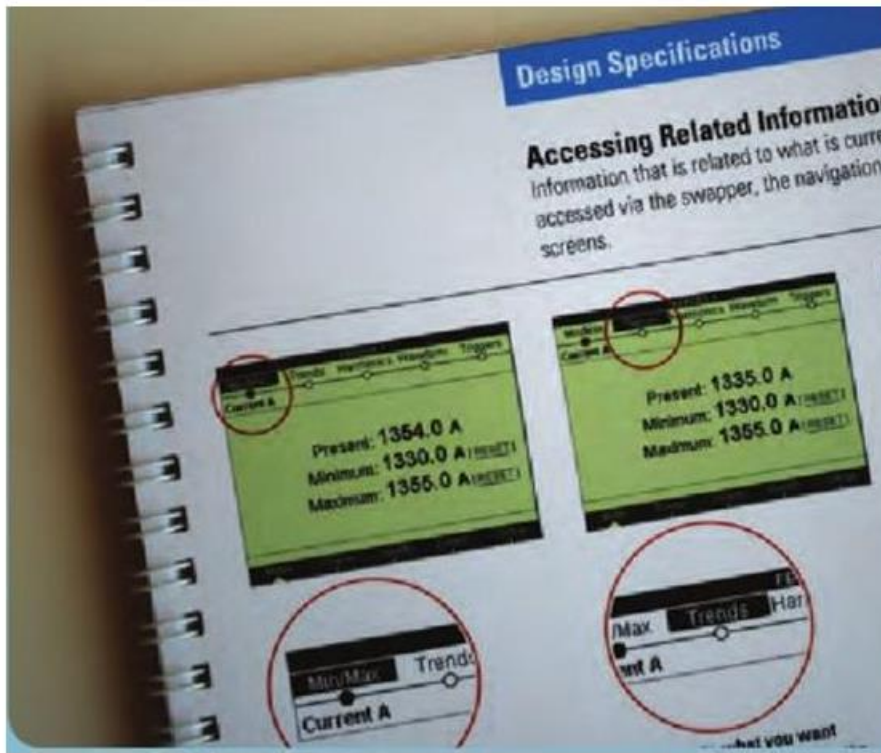
- Shows a successful future state
- Promotes a shared vision
- Gains support from decision makers
- Inspires your team

- Identify a new concept to promote.
- Assemble a team and some basic drawing materials.
- Select a relevant magazine and draw the masthead.
- Date the magazine as if it is a future issue.
- Write a newsworthy headline and a subheading.
- Draw an illustration for the cover artwork.
- Write the first paragraph of the story.
- Draw supporting illustrations for the inside spread.
- Include callouts, quotes, and sidebar stories.

Quick Reference Guide

- A short document summarizing the key principles and elements of a proposed solution.
- A Quick Reference Guide ensures that your original design intent is maintained as the idea moves from conception through production and beyond. Since clarity is the goal, a good guide is essential.
- Everything relevant to the design—style, form, composition, and conventions—is contained in one document that provides an orientation for those unfamiliar with the concept. Therefore, it must be easy to digest.
- Begin your Quick Reference Guide as early in the process as you can.
- This will offer you a chance to clearly articulate your team's design rationale as soon as you begin making decisions.
- Then the document can evolve along with the project, eventually containing all of the information that defines the design.
- When completed the document should answer these questions: What is it? What are the elements? How does it work?

This example highlights a page of design guidelines for producing a new piece of equipment. The development team used it as a quick reference for key specifications.



BENEFITS

- Summarizes the rationale for your design
 - Promotes key principles
 - Provides specifications
 - Supports proper implementation
- Identify a design that is ready to be documented.
 - Assemble the team that originated the solution.
 - Ask them to communicate the design intent.
 - Summarize the guiding principles for the design.
 - Indicate the standards for color, typography, etc.
 - Provide a library of design patterns and templates.
 - Package the specifications in a concise manner.
 - Prepare a small booklet and/or an online reference.

Design Vs Development

- The term “**design**” is used broadly to refer to the **entire system development lifecycle**.
- It is also used narrowly to describe the creative synthesis of ideas in interaction design.
- Involves understanding the audience and creating a design that meets their needs.
- Involves creating and testing prototypes.
- “**Develop**” and “**development**” could work but are strongly associated with software engineering and coding.
- Involves the physical creation of a product or service.
- Involves translating a prototype into reality.

DESIGN PARADIGMS

- A design paradigm is a way of thinking about and approaching design.
- It's a set of ideas, methods, and assumptions that guide how designers solve problems.
- Different design paradigms focus on different aspects of human-computer interaction (HCI), such as efficiency, cognition, or meaning-making.
- Paradigms shape how designers understand users, create interfaces, and evaluate systems.
- The different design paradigms are:
 - Engineering Paradigm
 - Human Information Processing (HIP) Paradigm
 - Design-Thinking Paradigm

1. Engineering Paradigm

- The HCI Engineering Paradigm, drawing inspiration from software engineering and human factors, **emphasizes a structured approach to designing user-friendly systems.**
- **Begins by defining the desired system functionalities.**
- **Prioritizes usability engineering, involving continuous evaluation and improvement to enhance user performance.**
- Employs a structured design process with **regular evaluations and refinements.**
- Leverages human factors research to understand user needs, behaviors, and limitations.
- **Focuses on fulfilling user requirements and maximizing productivity.**
- **Prioritizes efficiency and error reduction,** often measured by user performance metrics.

- An example is the study of an assembly line where each action required to do work efficiently was described carefully.
- These descriptions were then more or less translated into requirements.
- Designs focused on how to support these requirements and to automate where desirable.
- It was a purely utilitarian and requirements-driven approach.
- Success was measured by how much the user could accomplish, and alternative methods and designs were compared with statistical summative studies.

2. Human Information Processing (HIP) Paradigm

- The Human Information Processing (HIP) paradigm in HCI **views the human mind and the computer as interconnected information processors.**
- HIP emphasizes **understanding how humans perceive, process, and act upon information** (sensing, cognition, memory, decision-making, and physical actions).
- By modeling human cognitive processes, designers can create interfaces that align with human capabilities and limitations.

- The HIP paradigm has its roots in psychology and human factors, from which it gets an element of cognitive theory.
- It is about modeling human sensing, cognition, memory, information understanding, decision making, and physical performance in task execution.
- The idea was that once these human parameters were codified, it is possible to design a product that “matches” them.
- Example: The 7 +/- 2 rule, which suggests limiting the number of items in a list due to the constraints of human short-term memory, is a classic example of applying HIP principles. (In the context of HCI this means that menus, toolbars, or any other interface elements that present a list of options should ideally have no more than 7 items.)

3. Design-Thinking Paradigm

- Also called the “phenomenological matrix.”
- The Design-Thinking Paradigm is about designing for the experience, not just the product itself.
- It considers the social, cultural, and emotional aspects of interaction.
- Moves beyond traditional usability methods, which focus on refining existing designs rather than creating the right design from the start.
- Uses participatory design to involve users in early prototypes and design exploration.

- Shifts from evaluating single-task performance to considering emergent use and overlapping design phases (analysis, design, implementation, evaluation).
- Focuses on emotional, cultural, and social aspects of interaction, incorporating ideas from psychology, anthropology, and technology.
- Considers "**embodied interaction**", where interaction involves the whole body and spirit, not just traditional input devices.
- "**Situated design**" emphasizes the role of place in human-technology interaction.
- Promotes joy, aesthetics, and a deep connection to products beyond mere functionality.

Real-World Application: The iPad Example

- Critics argued the iPad lacked features (e.g., note-taking) and utility (replacing laptops/desktops).
- However, the iPad's success lies in the experience: a beautiful device for consuming information in a new way.
- Before the iPad, there were email, digital newspapers such as CNN.com, book readers, and photo viewers, but this device introduced an experience in doing these same things that was unprecedented.
- In summary, the design-thinking paradigm shifts HCI focus from just utility and usability to user experience, emotions, and cultural significance, creating designs that deeply resonate with people.

Car Design Example to Contrast Paradigms:

- **Engineering Paradigm:** Focuses on utility, performance, and mechanics (e.g., horsepower, fuel efficiency).
- **Usability Paradigm:** Improves ergonomics, safety, and control accessibility.
- **Design-Thinking Paradigm:** Prioritizes emotional appeal, the joy of driving, and the personal connection to the car.

DESIGN THINKING

- Design thinking shifts the focus from simply building a functional product to creating a product that is both useful and emotionally engaging, making it a key driver of success in today's competitive market.
- Traditional View: Design is often treated as a cosmetic afterthought, focused solely on aesthetics.
- **Design Thinking**: prioritizes user experience, emotions, and aesthetics from the very beginning.
- Is a holistic approach, considering user needs, market trends, and the overall brand experience.
- Emphasizes creativity, empathy, and a deep understanding of human needs.
- Drives product development, not just follows it.
- Example: **The iPod Touch**: Illustrates the power of design thinking. Its success stems not just from functionality, but also from its beautiful design, intuitive interface, and the overall user experience.

DESIGN PERSPECTIVES

- We describe three design perspectives as filters through which we view design and design representations to guide thinking, scoping, discussing, and doing design.
 - Ecological Perspective
 - Interaction Perspective
 - Emotional Perspective

Ecological Perspective

- The ecological design perspective is about **how the system or product works within its external environment.**
- It is about how the system or product is used in its context and how the system or product interacts or communicates with its environment in the process.
- This is a work role and workflow view, which includes social interaction and long-term phenomenological aspects of usage as part of one's lifestyle.
- **System infrastructure plays an important role in the ecological perspective** because the infrastructure of a system, the other systems and devices with which it interacts in the world, is a major part of its ecology.
- Infrastructure leads you to think of user activities, not just isolated usage.
- This perspective **shifts the focus from simply producing a physical object to designing a product that provides a valuable service within a larger system, considering its environmental and social impact throughout its lifecycle.** - **“A product is actually a service.”**

Emotional Perspective

- The emotional design perspective is about **emotional impact and value-sensitive aspects of design**.
- It is about **social and cultural implications**, as well as the aesthetics and joy of use.
- **System infrastructure can also play a role in the emotional perspective** because the infrastructure of a system provides scaffolding for the phenomenological aspects of usage, which are about broader usage contexts over longer periods of time.
- **A product is not just a product; it is an experience.**
- People do not usually use a product in isolation from other activities.
- **People use products as part of an activity**, which can include many different kinds of usage of many different things.
- And that starts with the **out-of-the-box experience**, which is not enhanced by difficult hard plastic encasing, large user manuals, complex installation procedures, and having to consent to a legal agreement that you cannot possibly read.

Interaction Perspective

- The interaction design perspective is about **how users operate the system or product.**
- It is **a task and intention view, where user and system come together.**
- It is where users look at displays and manipulate controls, doing sensory, cognitive, and physical actions.

USER PERSONAS

- Persona is a story and description of a specific individual who has a name, a life, and a personality.
- Personas are not real users but rather archetypal users that embody the characteristics, needs, and goals of actual users.
- They are built from contextual research and data, providing a vivid picture of the target audience.
- A persona typically has:
 - A name
 - A background story
 - Specific behaviors and motivations

Why Personas Are Used:

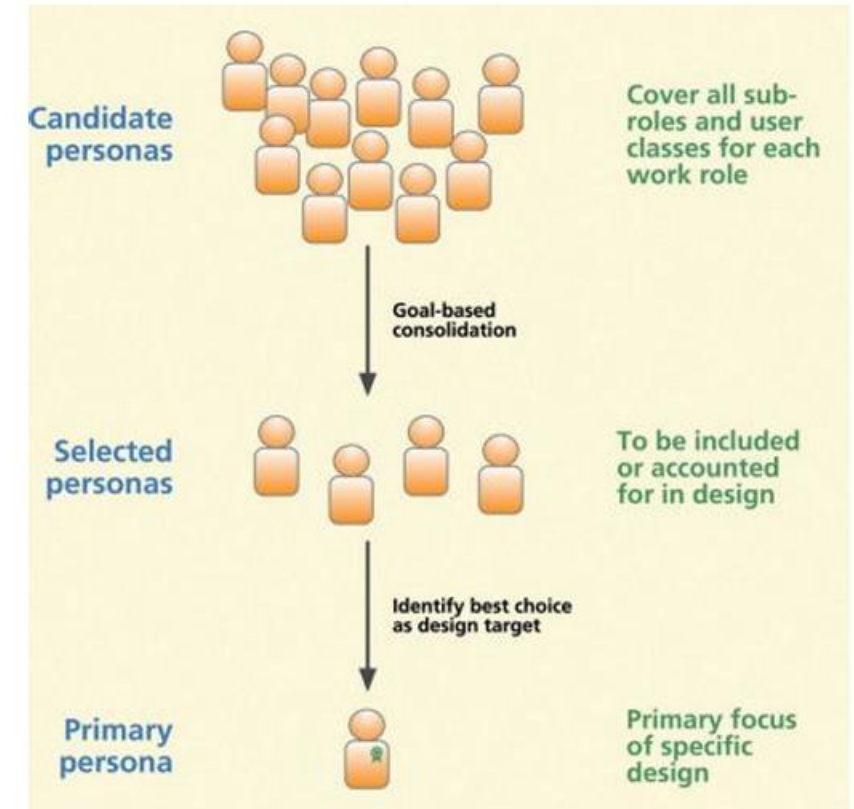
- **Avoid unnecessary features:** Personas help designers focus on what's truly important for the target user and avoid cramming in unnecessary features.
- **Resolve design conflicts:** When faced with design decisions, personas provide a clear reference point to determine the best course of action for the target user.
- **Design for specific needs:** Personas help designers consider the specific needs, goals, and behaviors of different user types, leading to more targeted and effective designs.
- **Avoid designing for yourself:** Personas prevent designers from unconsciously designing for themselves or similar users, ensuring the product caters to the actual target audience.

Mechanics of Creating Personas

- Each persona should have:
 - A first and last name (fictional but realistic)
 - A photograph or illustration
 - A work role and context
 - Goals, pain points, and motivations

How Personas Are Created:

- **Identify candidate personas:** Based on user research (interviews, surveys), identify potential user groups with distinct characteristics and goals.
- **Goal-based consolidation:** Merge personas with overlapping goals and characteristics to reduce complexity.
- **Select a primary persona:** Choose the persona that best represents the core user group and design the product primarily for them.



Characteristics of Effective Personas

- **Make your personas rich, relevant, believable, specific, and precise:** A persona should have a backstory, personality, and specific details that make them believable and relatable.
- **Make your personas “sticky”:** Personas should be easy to remember and reference throughout the design process.
- **Relevant and consistent:** The persona should represent a realistic user type within the target audience.

IDEATION

- Ideation is an active, fast-moving collaborative group process for forming ideas for design.
- It is an activity that goes with design thinking; ideation is a tool of design thinking.
- **Iterative Exploration**: Ideation is about exploration—it requires constant iteration.
 - Designers must test multiple ideas, similar to how Thomas Edison tested thousands of materials before perfecting the lightbulb.
 - Sketching and mockups should be made early and often, allowing users to be involved in refining the design.
- Idea Creation vs. Critiquing
 - There are two main modes of thinking in ideation:
 - i. **Idea Creation** → Generating new ideas freely, brainstorming without restrictions.
 - ii. **Critiquing** → Evaluating and refining ideas based on feasibility and effectiveness.

Doing Ideation

1. Setting Up Workspaces:

- Establish dedicated spaces for ideation, individual work, and group collaboration.
- If possible, create an ideation studio like the Virginia Tech “Kiva,” designed to minimize distractions and maximize creativity.

Fig: The Virginia
Tech ideation
studio, the “Kiva”



Fig: Individual and group
designer work spaces



2. Assembling a Team:

- Creativity flourishes in teams, not just from lone geniuses.
- Include people from diverse disciplines and backgrounds, along with customer representatives.
- Encourage open-mindedness and freedom to generate ideas without fear of criticism.

3. Use ideation bin ideas to get started:

- Gather work activity notes and emotional impact factors from prior research.
- Organize, group, and label them as prompts for brainstorming discussions.

4. Brainstorming Process:

- Begin with a group overview to align goals and expectations.
- Break into smaller groups for intensive idea generation using sketches and flip charts.
- Reconvene to share, discuss, and expand upon ideas.
- Maintain a balance of verbal and visual thinking.
- Critique should focus on improvement, not rejection.



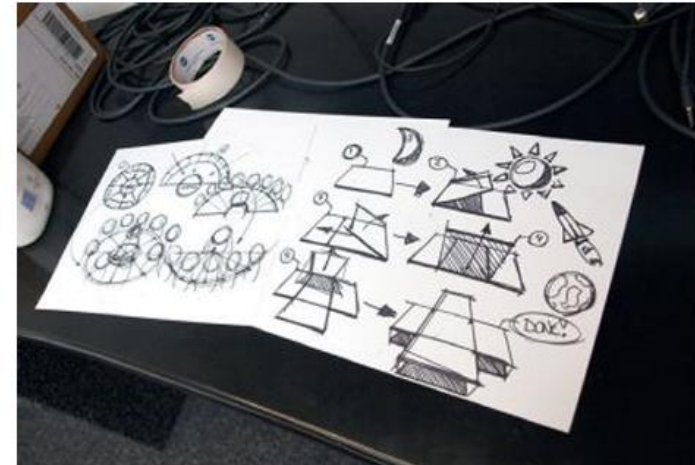
SKETCHING

- Sketching is a **rapid, freehand method for generating and exploring design ideas.**
- Sketching for Ideation: Sketching **enhances creativity** by providing a visual language for brainstorming, **helping designers explore possibilities** rather than refine finalized ideas.
- Sketching vs. Prototyping: Unlike prototypes, which define and test decisions, sketches are ambiguous, tentative, and **used for idea generation.**
- Embodied Cognition: Sketching is not just a representation of thought—it is an **active part of thinking and problem-solving, engaging both mind and body.**

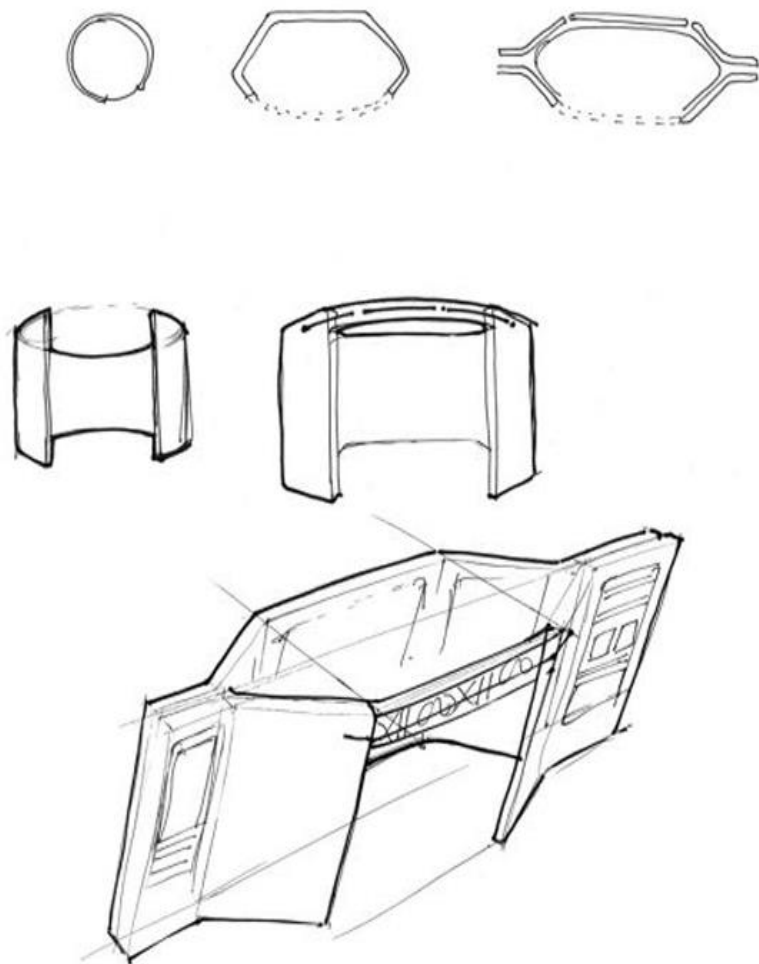
- Everyone can sketch; you do not have to be artistic.
- Most ideas are conveyed more effectively with a sketch than with words.
- Sketches are quick and inexpensive to create; they do not inhibit early exploration.
- Sketches are disposable; there is no real investment in the sketch itself.
- Sketches are timely; they can be made just-in-time, done in-the-moment, provided when needed.
- Sketches should be plentiful; entertain a large number of ideas and make multiple sketches of each idea.
- Textual annotations play an essential support role, explaining what is going on in each part of the sketch and how.

How Sketching Helps in Design Thinking

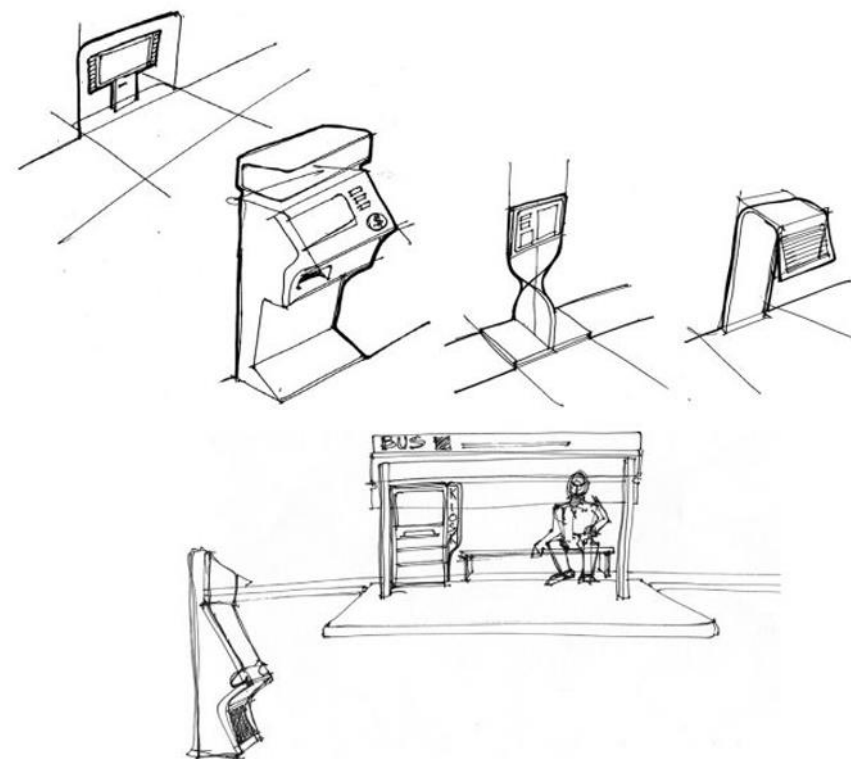
- **Encourages Quick Idea Exploration:** Sketches enable rapid iteration and brainstorming.
- **Supports Collaboration:** Teams can annotate, modify, and discuss sketches together.
- **Serves as Design Documentation:** Sketches preserve the evolution of ideas throughout a project.
- **Example:** A sketch of an app's navigation flow helps designers visualize user interactions before coding begins.



Freehand
gestural
sketches
for the
Ticket Kiosk
System



Ideation and
design exploration
sketches for the
Ticket Kiosk System

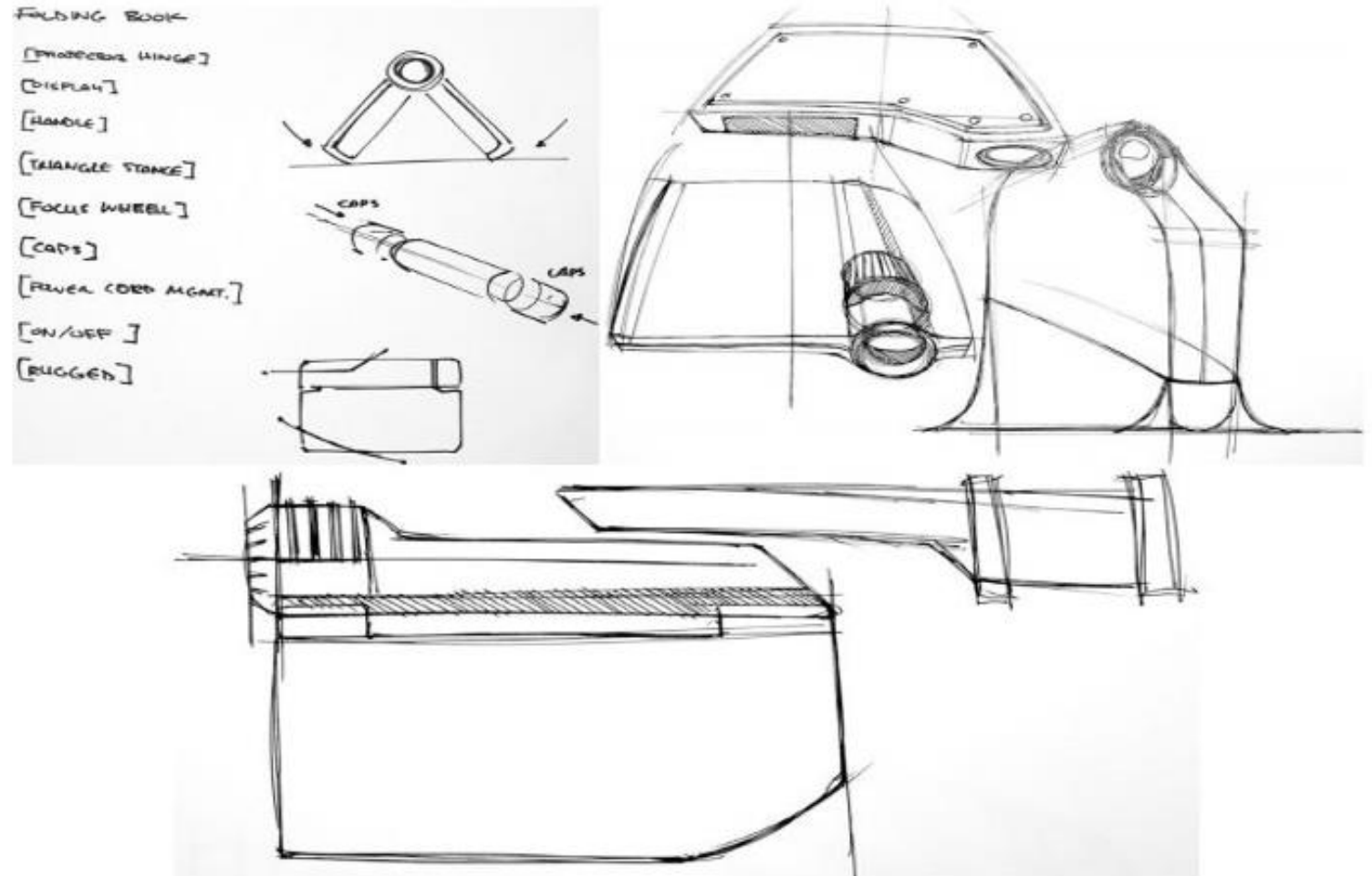


Example: Sketching for a Laptop/Projector Project

The following figures show sample sketches for the K-YAN project (K-yan means “vehicle for knowledge”), a collaboration by the Virginia Tech Industrial Design Department.

The objective is to develop a combination laptop and projector in a single portable device for use in rural India.

Fig1: Early ideation sketches of K-YAN



Physical Mockups

- Physical mockups serve as **three-dimensional sketches**, allowing designers to explore **tangible interactions quickly and cheaply**.
- These are especially **useful for product design and usability testing**.

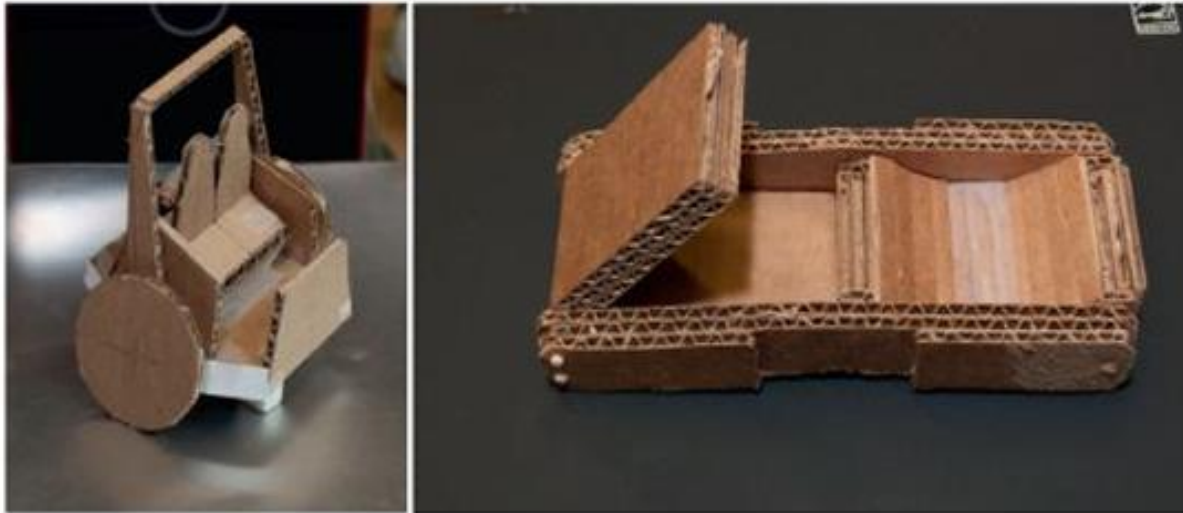


Fig1: Examples of rough physical mockups



Fig2: Example of a more finished looking physical mockup